HTML + CSS

Building and Styling the Web

CSCI 39548 — Practical Web Development

Section Goals

- Understand what HTML is and how browsers read it
- Write well-structured HTML documents
- Style pages with CSS (inline, internal, external)
- Use Flexbox and Grid for layout
- Build responsive, polished pages with transitions

Section Items

- HTML Primitives
- Lists and Tables
- Semantic HTML
- Forms and Inputs
- CSS — Inline / Internal / External
- Selectors and the Box Model
- Typography, Units, and CSS Variables
- Flexbox
- Grid and Positioning
- Responsive Design and Transitions

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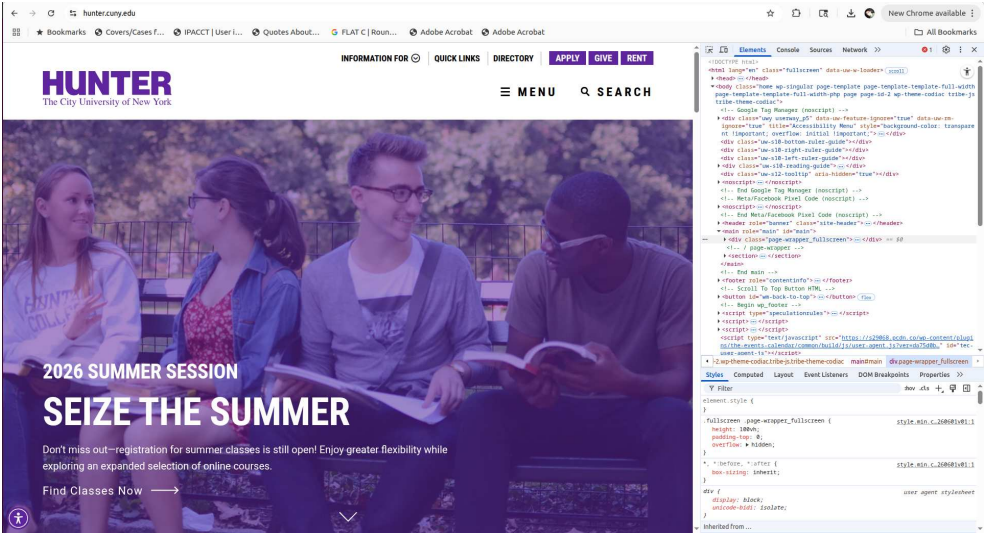
HTML — The Structure of the Web

- HTML = HyperText Markup Language
- Describes the **structure** of a page, not how it looks
- An HTML file is just plain text — the browser reads and renders it
- Every website you've ever visited is HTML at its core

Every Page is HTML



Tags and Attributes

```
<a href="https://example.com" target="_blank">Visit</a>

<img src="photo.jpg" width="300">

<!-- This is a comment -->
```

- Tags wrap content: opening and closing
- Some are self-closing: <img>,
, <hr>
- **Attributes** add info: `href`, `src`, `target`, `width`
- Comments: <!-- ignored by browser -->

The Document Skeleton

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>My Page</title>
</head>
<body>
    <h1>Hello World</h1>
</body>
</html>
```

- `<!DOCTYPE html>` — tells browser this is HTML5
- `<head>` — metadata (not visible on the page)
- `<body>` — all visible content goes here

Common HTML Tags

Tag	Purpose
`<h1>` – `<h6>`	Headings (h1 biggest, h6 smallest)
`<p>`	Paragraph
`<a href="...">`	Link
`<img src="...">`	Image
`<button>`	Clickable button
`<hr>`	Horizontal rule (divider)
` `	Line break

Live Code: snapshot-01

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I teach Practical Web Development at Hunter College, CUNY. I love building things, breaking them, and explaining how they work.

About

Based in New York. I write code, I teach code, and sometimes I do both at the same time. This portfolio is also a teaching demo for my CSCI 39548 students.

Contact

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GitHub Profile
LinkedIn

Download Resume

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Lists and Tables

Structured Content in HTML

Unordered vs Ordered Lists

```
<ul>
    <li>Milk</li>
    <li>Eggs</li>
</ul>

<ol>
    <li>Preheat oven</li>
    <li>Mix ingredients</li>
</ol>
```

- `<ul>` — bullets, order doesn't matter
- `<ol>` — numbered, order matters
- `<li>` — list item (used inside both)

Lists in the Wild

HTML Lists

Unordered list (<ul>)

- Apples
- Bananas
- Cherries

Ordered list (<ol>)

1. Wake up
2. Make coffee
3. Write code
4. Take a break

Nested list

- Frontend
  - HTML
  - CSS
  - JavaScript
    - React
    - Vue
- Backend
  - Node.js
  - Python

Description list (<dl>)

HTML
: The structure of a web page.

CSS
: The visual styling of a web page.

JavaScript
: The behavior and interactivity of a web page.

Nested Lists

```
<ul>
    <li>Frontend
        <ul>
            <li>HTML</li>
            <li>CSS</li>
            <li>JavaScript</li>
        </ul>
    </li>
    <li>Backend</li>
</ul>
```

- Lists can nest inside other lists
- Inner `<ul>` goes inside an `<li>`

Description Lists

```
<dl>
    <dt>HTML</dt>
    <dd>Structure of the page</dd>
    <dt>CSS</dt>
    <dd>Styling and layout</dd>
</dl>
```

- `<dl>` — description list
- `<dt>` — term, `<dd>` — definition
- Rare but valid — good for glossaries

Table Structure

```
<table>
    <thead>
        <tr>
            <th>Name</th>
            <th>Grade</th>
        </tr>
    </thead>
    <tbody>
        <tr>
            <td>Alice</td>
            <td>A</td>
        </tr>
    </tbody>
</table>
```

Tables in the Wild

HTML Tables

Minimal table

Apple	Red
Banana	Yellow
Grape	Purple

Proper table structure

Course	Instructor	Credits	Semester
CSCI 39548 - Practical Web Development	Sourya Saha	3	Summer 2026
CSCI 41500 - Data Communications and Networking	Sourya Saha	3	Spring 2026
CSCI 235 - Software Design	TBD	3	Fall 2026

Merged cells (colspan / rowspan)

Day	Morning	Afternoon
Monday	Office hours	CSCI 39548 lecture (6:40 - 9:48 PM)
Wednesday	Research	
Weekend: catch up on grading		

Spanning Cells

```
<td colspan="2">Spans two columns</td>

<td rowspan="3">Spans three rows</td>
```

- `colspan` — merge across columns
- `rowspan` — merge across rows

Tables are NOT for Layout

- In the 1990s, developers used `<table>` for page layout
- That era is over — use Flexbox and Grid (covered later)
- Tables are **only** for displaying tabular data
- If your content isn't rows and columns of data, don't use a table

Live Code: snapshot-02

- Let's open `snapshot-02/lists.html` and `tables.html`
- Walk through the list types, nesting, and table structure

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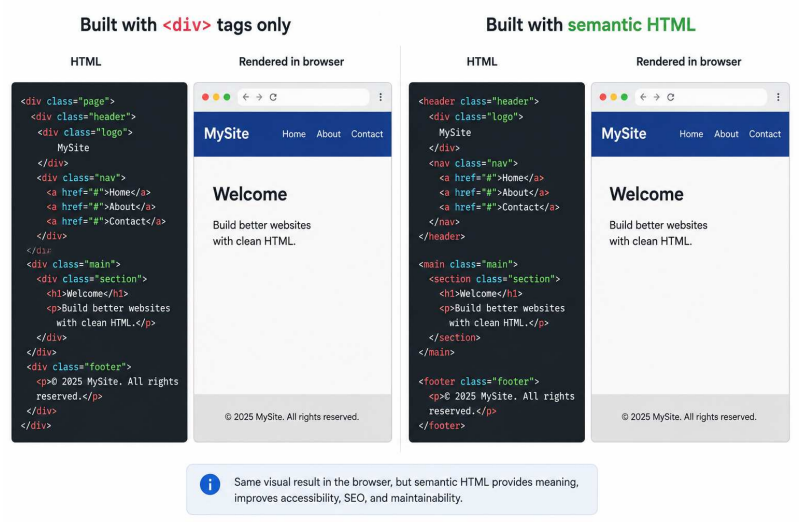
Semantic HTML

Meaning Over Markup

The Problem with div Everywhere

- You could build an entire page with only `<div>` tags
- It would render fine — but no one (including machines) knows what anything means
- Screen readers can't navigate
- Search engines can't prioritize
- Your future self can't read the code

div vs Semantic Tags



The Semantic Landmark Tags

- `<header>` — top of page (logo, title, nav)
- `<nav>` — navigation links
- `<main>` — primary content (exactly ONE per page)
- `<section>` — thematic chunk with its own heading
- `<article>` — self-contained piece (blog post, card)
- `<aside>` — tangentially related (sidebar, callout)
- `<footer>` — bottom of page (copyright, links)

Why Semantic Tags Matter: Accessibility

- Screen readers announce landmarks: "navigation," "main content," "footer"
- Blind users can jump directly to `<main>` or `<nav>`
- `<div>` is invisible to assistive technology
- Semantic HTML is the foundation of accessible web development

Why Semantic Tags Matter: SEO

- Google reads `<main>` first to understand your page's core content
- `<nav>` helps crawlers find your site's link structure
- `<article>` signals standalone content worth indexing
- Better semantics $\rightarrow$ better search ranking

section vs article

- `<section>` = a chapter inside a larger whole (can't stand alone)
- `<article>` = could be lifted out and still make complete sense
- The test: "Would this make sense in an RSS feed or shared on its own?"
  - Yes → `<article>`
  - No → `<section>`

The id Attribute and Anchor Links

```
<nav>
    <a href="#about">About</a>
    <a href="#projects">Projects</a>
</nav>

<section id="about">
    <h2>About Me</h2>
</section>
```

- `id` = unique identifier on an element
- `href="#about"` jumps to the element with `id="about"`
- Also used by CSS and JavaScript selectors

Live Code: snapshot-03

- Let's open `snapshot-03/index.html`
- Click the nav links — watch the page jump to sections